Analysis of Myntra Apparel

Case Study

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Tools Used: Microsoft Excel

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A. Data Cleaning and Preparation

Checked the dataset for duplicate records using Excel. No duplicate rows were found. Therefore, no records were removed.

Total Rows Before Removing Duplicates	5,26,564
Duplicate Rows Found	0
Total Rows After Removing Duplicates	5,26,564

Standardized the DiscountOffer column

Removed unnecessary text such as **"OFF"**, **"Rs."**, *", *Hurry**", and extra spaces to standardize the DiscountOffer column.

The numeric value was extracted from each entry, resulting in a consistent numeric format suitable for analysis.

✓ Before Cleaning the DiscountOffer Column	✓ After Cleaning the DiscountOffer Column
40% OFF	40
70 % OFF	70
55% OFF , Hurry*	55
Rs. 1100 OFF	1100

Function Used to Clean the DiscountOffer column:

```
=SUBSTITUTE(
SUBSTITUTE(
SUBSTITUTE([@DiscountOffer],
"OFF",""),
"Rs.",""),
", Hurry*","")
```

The cleaned values included both percentage and fixed-amount discounts. The NUMBERVALUE() function converted the extracted text into numeric values for further analysis.

Function Used :-

```
=NUMBERVALUE(SUBSTITUTE( SUBSTITUTE( SUBSTITUTE([@DiscountOffer], "OFF",""), "Rs.",""), ", Hurry*",""))
```

Converted all discount values into percentage discounts to determine the actual discount percentage for each product. For fixed-amount discounts (e.g., ₹1100 OFF), the percentage discount was calculated using the Original Price. Percentage-based discounts were retained without modification. This ensured all discount values were represented in a single, consistent percentage format for accurate analysis.

Function Used:

=(IF(NUMBERVALUE(SUBSTITUTE(SUBSTITUTE(SUBSTITUTE([@DiscountOffer],"OFF",""),"Rs.",""),", Hurry*",""))>0,

IF(NUMBERVALUE(SUBSTITUTE(SUBSTITUTE(SUBSTITUTE([@DiscountOffer],"OFF",""),"Rs.",""),", Hurry*",""))<1,

NUMBERVALUE(SUBSTITUTE(SUBSTITUTE(SUBSTITUTE([@DiscountOffer],"OFF",""),"Rs.",""),", Hurry*",""))*[@[OriginalPrice (in Rs)]],

NUMBERVALUE(SUBSTITUTE(SUBSTITUTE(SUBSTITUTE([@DiscountOffer],"OFF",""),"Rs.",""),", Hurry*",""))),

NUMBERVALUE(SUBSTITUTE(SUBSTITUTE(SUBSTITUTE([@DiscountOffer],"OFF",""),"Rs.",""),", Hurry*",""))))/[@[OriginalPrice (in Rs)]])*100.

Functions Used:-

- SUBSTITUTE()
- NUMBERVALUE()
- Nested IF()

Result:- The DiscountOffer column was successfully standardized into numeric values, making it suitable for calculations, filtering, and analysis.

1. Handled Missing DiscountPrice Values

- Some products had missing values in both the **DiscountPrice** and **DiscountOffer** columns. Since no discount information was available for these products, the missing **DiscountPrice** values were imputed using the **average DiscountPrice of the respective product category**. This ensured that the records were retained for analysis while maintaining category-level pricing consistency.

Function Used

```
=IF(AND([@DiscountPrice]="",[@DiscountOffer]=""),
AVERAGE(FILTER([DiscountPrice],[Category]=[@Category])),
[@DiscountPrice])
```

Result:

- Filled missing values in the **DiscountPrice** column.
- Retained products with missing discount information.
- Preserved category-wise pricing patterns for further analysis.

Handled Missing SizeOption Values

- The **SizeOption** column contained blank values for products where size information was unavailable. All missing values were replaced with **"Not Available"** to maintain consistency and improve readability in reports and Pivot Tables.

Function Used

```
=IF([@SizeOption]="","Not Available",[@SizeOption])
```

SizeOption	Availabilit
S, M, L, XL, XXL	S, M, L, XL, XXL
XS, S, M, L, XL	XS, S, M, L, XL
	Not Available
30, 32, 34, 36	30, 32, 34, 36
	Not Available
XS, S, M, L, XL	XS, S, M, L, XL
28, 30, 32, 34, 36	28, 30, 32, 34, 3
	Not Available

Result:

- Eliminated blank values in the **SizeOption** column.
- Standardized missing entries.
- Improved filtering and reporting.

B. Data Analysis Using Excel Functions

1. Calculated the Average Original Price of Highly Rated Products

- To understand the pricing of well-rated products, the average **Original Price** was calculated for products with a **Rating greater than 4**.

Function Used:

=AVERAGEIF(P:P,">4",J:J)

OR

=AVERAGEIF([Rating],">4",[OriginalPrice (in Rs)])

Average Original Price: ₹1,966.67

P:P = Range Of Ratings

J:J = Range of Original Price

Result:

- Calculated the average original price of all products with ratings above 4.
- Helped analyze the pricing trend of highly rated products.

Insight: Products with ratings above 4 have an average original price of ₹1,966.67. This suggests that, within this dataset, highly rated products tend to be priced higher on average.

2. Counted Products with More Than 50% Discount

- The number of products offering a discount greater than **50%** was calculated to identify how many products were heavily discounted.

Function Used:

- =COUNTIF([DiscountPercentage],">50")

Products with Discount > 50%: 2,21,864

Result:

- Counted all products with discounts greater than 50%.
- Helped measure the proportion of highly discounted products.

Insight: Only 2,21,864 products offer discounts above 50%, suggesting that heavy discounting is limited to a small portion of the catalog.

3. Counted Products Available in Size "M"

- The **COUNTIF()** function was used to determine how many products were available in **Medium (M)** size.

Function Used:

- =COUNTIF([SizeOption],“*M*")
- **Products Available in Size M:3,08,460**

Result:

- Counted all products available in Medium size.
- Assisted in analysing size availability.

Insight: Medium (M) is one of the most frequently available sizes, indicating strong inventory focus for this size category.

4. Categorized Products Based on Discount Percentage

- A new column named **Discount Category** was created to classify products according to their discount percentage.
- **High Discount** → Discount greater than 50%
- **Low Discount** → Discount 50% or less

Formula Used:

- =IF([@DiscountPercentage]>50,"High Discount","Low Discount")

55	High Discount
55	High Discount
31	Low Discount
35	Low Discount
40	Low Discount
60	High Discount
58	High Discount
49	Low Discount
49	Low Discount
55	High Discount
49	Low Discount
70	High Discount
50	High Discount
60	High Discount
53	High Discount
59	High Discount

Result:

- Products were categorized into High Discount and Low Discount groups.
- This classification was later used for filtering, Pivot Tables, and visualization.

C. Data Retrieval & Lookup Operations

1. Retrieved Product Details Using VLOOKUP/XLOOKUP

- The **VLOOKUP()** and **XLOOKUP()** functions were used to retrieve the **Brand**, **Original Price**, and **Rating** of the product with **Product ID 11226634**. This demonstrates how lookup functions can efficiently fetch related information from large datasets using a unique identifier.

Function Used:-

=VLOOKUP("11226634",Table1,Column_Number,FALSE)

OR

=XLOOKUP(11226634,Product_ID,Brand)

Example:- for Brand Name of Product Id 11226634

=VLOOKUP(11226634,B2:Q526565,2,FALSE)

Output:-Maniac

Result:

- Successfully retrieved the Brand, Original Price, and Rating of the selected product.
- Demonstrated the use of lookup functions for efficient data retrieval.

2. Retrieved DiscountPrice Using INDEX and MATCH

- The **INDEX()** and **MATCH()** functions were combined to retrieve the **DiscountPrice** of the product with **Product ID 6744434.** This method provides a flexible alternative to VLOOKUP, especially when the lookup column is not the first column in the dataset.

Formula Used:

=INDEX(E2:E526566,MATCH(6744434,B2:B526566,0))

Result:

- Successfully retrieved the DiscountPrice using INDEX-MATCH.
- Demonstrated a dynamic lookup approach suitable for large datasets.

3. Retrieved Product Information Using Nested XLOOKUP

- A **nested XLOOKUP()** formula was used to dynamically retrieve any product attribute by first identifying the product using its **Product ID** and then selecting the required column based on the user's input. This approach enables flexible retrieval of multiple fields using a single formula.

Formula Used:

=XLOOKUP(T2,

Table1[Product_ID],

XLOOKUP(U2,

Table1[#Headers],

Table1))

Result:

- Enabled dynamic retrieval of any product detail.
- Reduced the need for multiple lookup formulas.
- Improved flexibility and reusability of the analysis workbook.

Thank You
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